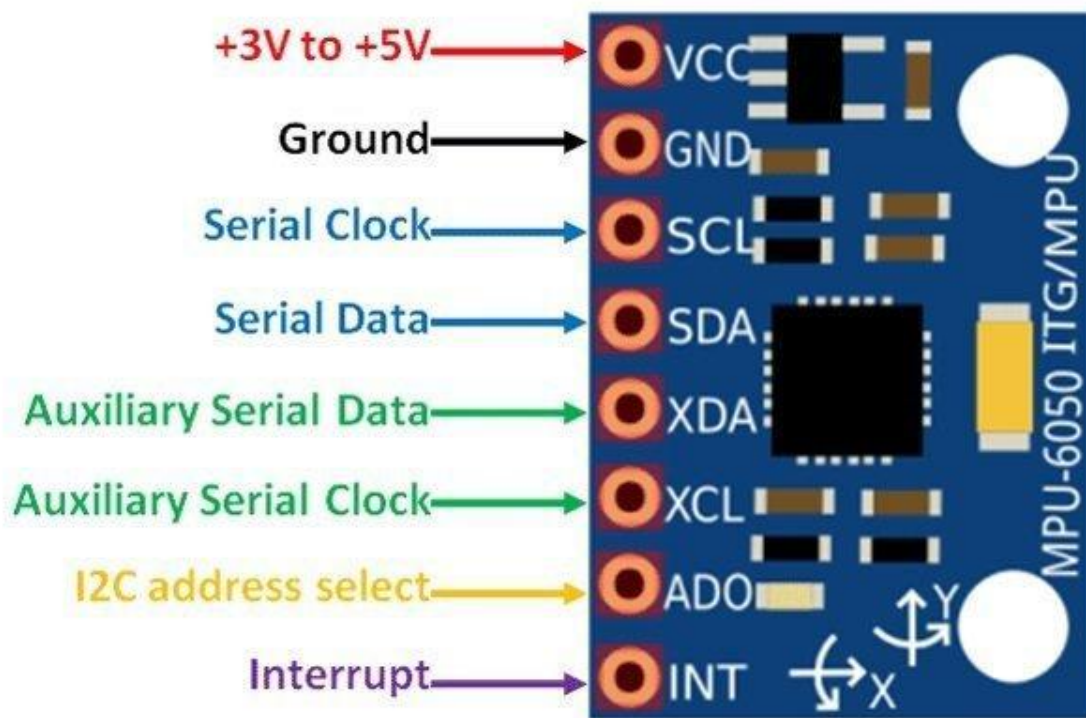


MPU6050

The **MPU6050** is a popular 6-axis motion tracking sensor that combines:

- **3-axis accelerometer** (measures acceleration and tilt)
- **3-axis gyroscope** (measures angular velocity/rotation)



MPU6050 Pinout

Accelerometer Definition

An accelerometer is a sensor that measures acceleration (change in velocity with time) and can also detect the tilt or orientation of an object with respect to gravity.

Accelerometer ek sensor hai jo kisi object ki movement, speed me hone wale badlav (acceleration) aur uske jhukav (tilt) ko measure karta hai.

Gyroscope Definition

A gyroscope is a sensor that measures the angular velocity (rate of rotation) of an object around one or more axes.

Gyroscope ek sensor hai jo kisi object ke ghoomne (rotation) ki speed aur direction ko X, Y aur Z axis par measure karta hai.

Digital Motion Processor

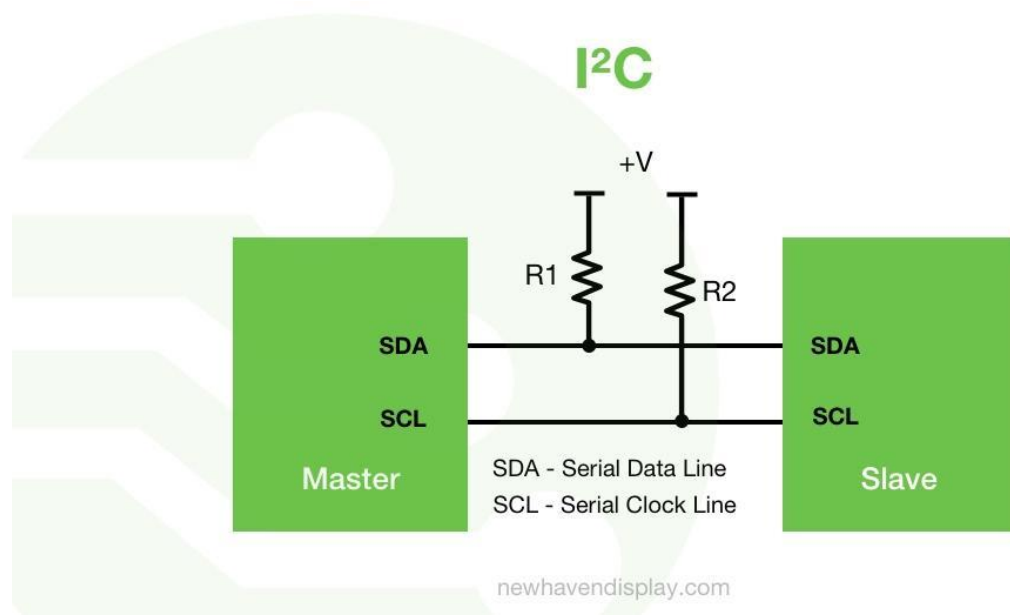
The DMP is an internal processor in the MPU6050 that combines accelerometer and gyroscope data to calculate accurate motion, orientation, and tilt information.

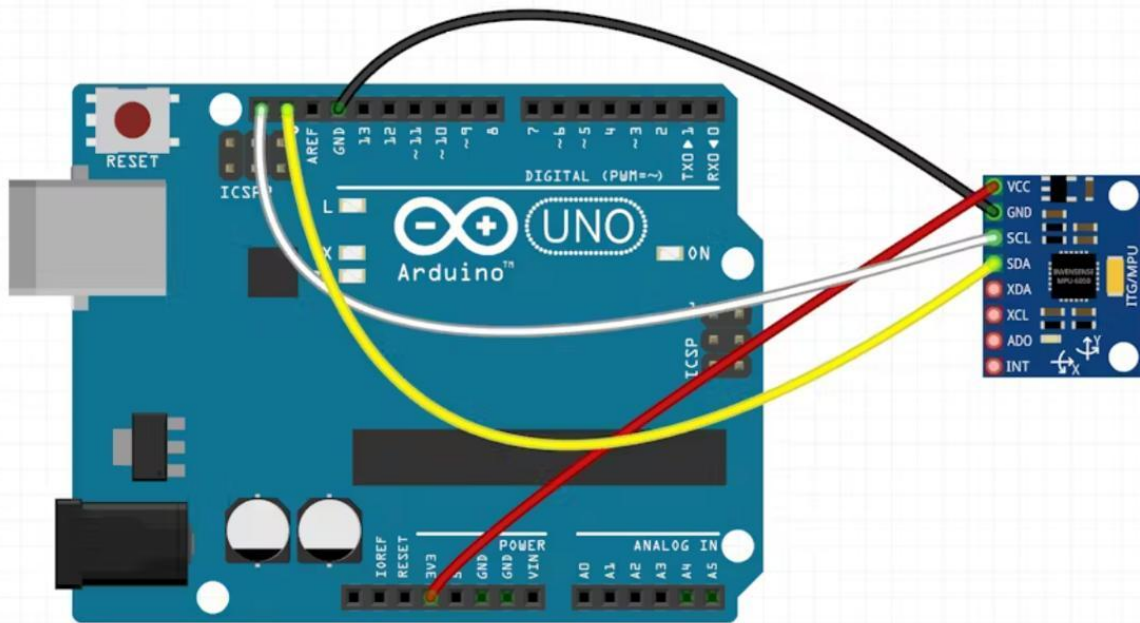
Key Specifications

- Supply voltage: 3V–5V (most modules)
- Communication: **I²C**
- Accelerometer ranges: $\pm 2g$, $\pm 4g$, $\pm 8g$, $\pm 16g$
- Gyroscope ranges: ± 250 , ± 500 , ± 1000 , ± 2000 $^{\circ}/s$
- Built-in Digital Motion Processor (DMP)

Applications

- Drones and quadcopters
- Self-balancing robots
- Mobile phones (gaming and motion sensing)
- Camera stabilization systems
- Navigation systems





```
//Mpu6050
#include<Wire.h>
#include<MPU6050_tockn.h>
MPU6050 mpu(Wire);
void setup()
{
  Serial.begin(9600);
  Wire.begin();
  mpu.begin();
  mpu.calcGyroOffsets(true);
}

void loop()
{
  mpu.update();
  Serial.println("Angle X :");
  Serial.println(mpu.getAngleX());
  Serial.println("Angle Y :");
  Serial.println(mpu.getAngleY());
  Serial.println("Angle Z :");
  Serial.println(mpu.getAngleZ());
  delay(50);
}
```

